EL NIÑO ACTION BRIEFING · 15 OCTOBER 2026

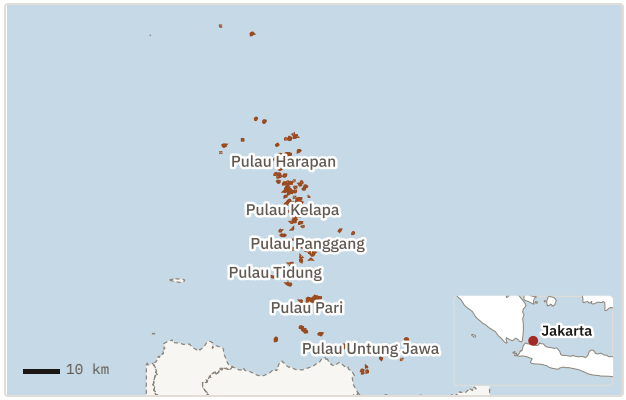
Kepulauan Seribu, Jakarta

On the Thousand Islands, El Niño arrives as dry wells beside desalination plants that cannot always keep up, and then as the west monsoon closes the sea. First, write the rule that moves sick people to hospital before the waves stop the boats, and keep every plant stocked with the parts it will need in January.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Jakarta Raya (whole province, one row)	~0.7 for Seribu, scaled (DKI: 260)	6-month, Sep 2026 – Feb 2027	Water shortfall now; the sea closing in Dec–Feb; dengue

15/15 recommendations checked. 1 confirmed as written, 6 corrected; 8 items of circulating advice contraindicated (the Do not cards). The funding target fell 94% in verification, from the panel's Rp656.5 m to Rp42.6 m: no published count of households without piped water, larvicide rounds sized by division, and fishers' insurance that protects income rather than lives. Exposure: Climate Impact Lab, *451,000 at Risk*, Sep 2026, pre-peer-review, heat only. Seribu's ~0.7 is our arithmetic, not an Impact Lab finding: 260 × 29,355 ÷ 10,669,702 = 0.72 (Dec–Feb: 147 × the same share = 0.40). Heat is not what kills here. Revised 15 October 2026 after the set audit; the verdicts are unchanged.

EXECUTIVE SUMMARY — AT A GLANCE



5.75°S 106.60°E, Indonesia · BPS boundaries via OCHA COD-AB; OpenStreetMap contributors (ODbL); Natural Earth

- **The risk.** Kelapa's wells ran dry in September; in January 2026 boats stopped twice in 1.25–2.5 m waves, and the only hospital is on Pramuka.
- **First action.** By 30 November, move high-risk patients when BMKG forecasts waves of 1.25 m or more, before the sea closes.
- **The money.** Rp22 m / Rp43 m / Rp119 m (US$1.2k / 2.4k / 6.7k); top placement: lidded drums for 112 Kelapa families.
- **Do not.** Do not sail aid when KSOP has withheld permits: the 2017 Zahro Express fire killed at least 23.

What to do	**7** actions	p. 4
Who does what	RACI chart	p. 11
Where money does the most	funding	p. 11
Do not do this	**8** items	p. 12
What has happened here	the record	p. 16
How this was produced	methodology	p. 18
References	sources	p. 19

* Suggested formulation: check with a health worker before sharing.

Kepulauan Seribu (Kabupaten Administrasi Kepulauan Seribu, Kemendagri 31.01) has 29,355 people (BPS, SP2020-based 2026 projection) on about a dozen inhabited islands, 15 to 80 km north of the Jakarta coast, in two kecamatan and six kelurahan: Pulau Panggang (with Pramuka, the regency seat), Pulau Kelapa, and Pulau Harapan in Kepulauan Seribu Utara; Pulau Untung Jawa, Pulau Tidung, and Pulau Pari (with Lancang) in Kepulauan Seribu Selatan. Its poverty rate, 12.37% (BPS, March 2025), is the highest in DKI. The bupati is appointed and the regency has no budget of its own, so this briefing names the regency offices (suku dinas) and the province bodies that do the work, and leaves province-wide items to the province file (`jakarta-province-*`). It is the second of six Jakarta briefings, written as a proposed draft (*draf usulan*) for the two-year review of the RPB 2023–2027 (Pergub DKI 1/2024).

The season. An El Niño dry season that is already here: it has dried shallow wells and the ends of the pipe network and held smog over the city; it is expected to delay and shorten the rains; and it does not make the January–February floods less likely. Each briefing therefore covers one season, from dry wells, smog, and heat now, through dengue bred in stored water and the floods, to March 2027.

How it lands on the islands. There is no mainland pipe here. Water comes from desalination plants that PAM Jaya runs, from shallow wells that turn brackish, from rain tanks, and by boat. The smog is the mainland's. What the islands get instead is the sea: the December–February west monsoon raises the waves that stop the boats carrying water, food, and patients, and the new-moon and full-moon tides push rob onto low ground.

WATER AND SEA STATUS	READING	DATE
BMKG drought early warning, Kepulauan Seribu (bmkg.go.id)	**Awas** (highest level); also at *Awas* for 11–20 Sep	Dasarian II, Oct 2026
Supply against need (regency and PAM Jaya)	About 1.45 M L a day produced against about 1.8 M needed; PAM Jaya 18.5 L/s to about 4,397 homes	11 Oct 2025 (latest found)
Plant faults on record	Pulau Kelapa seawater unit malfunctioning, bathing water short; no repair report found	4 Jul 2025 (latest found)
Water by boat (reported, not complete)	PHE OSES, 13,000 L to 112 families on Pulau Kelapa, wells dry or brackish for about six weeks	26 Sep 2026

* Suggested formulation: check with a health worker before sharing.

WATER AND SEA STATUS	READING	DATE
Waves, BMKG Maritim (Perairan Kep. Seribu Bagian Utara and Selatan)	No reading copied here: read maritim.bmkg.go.id daily. Last high-wave suspension: all four fast-boat routes, 1.25–2.5 m	28 Jan 2026
Coral heat stress, NOAA Coral Reef Watch "Northern Java" station	**Bleaching Watch**, degree heating weeks 0; station centred about 300 km east, coverage of Seribu unconfirmed	14 Oct 2026

BMKG put West, Central, East, and North Jakarta and Kepulauan Seribu at *Awas* for 11–20 September (BPBD DKI, via Kompas TV, 14 Sep); South Jakarta was not listed. BMKG's September bulletin is no longer online, and one report differs. BMKG's own page has Seribu at *Awas* again in dasarian II of October. Dengue by kecamatan (Dinkes DKI via Satu Data Jakarta): suspected cases in 2024, Kepulauan Seribu Utara 18 and Selatan 16; incidence in 2025 per 100,000, Selatan 118.54 and Utara 91.13. People: 6.81% are 65 or older (BPS, 2024); hypertension 30.81%, diabetes 2.97% (Riskesdas 2018).

THE ISLANDS' WATER

NAMED SECTION

PAM Jaya has been assigned the islands' seawater desalination (SWRO) plants since Pergub 62/2019, and by late 2025 the head of Sudin SDA Kepulauan Seribu, Mustajab, gave clean-water supply as fully shifted to PAM Jaya ("sepenuhnya dialihkan kepada PAM Jaya"); his office's budget fell from Rp201 bn in 2025 to Rp79 bn in 2026, mostly maintenance (Metro TV, 30 November 2025). Islanders pay a subsidised tariff (Rp3,550/m³ in 2021, against about Rp32,000 unsubsidised). The latest plant-by-plant list is from 2021 (JaKita): Kelapa and Harapan share 5 L/s; Panggang 3; Tidung 3; Lancang 1.5; Pramuka 1.5; Payung 0.25; Kelapa Dua 0.25. Untung Jawa's plant is reported at 2.5 L/s. Nothing published says which of them is running today. Brackish-water (BWRO) units at Panggang and Kelapa were planned for 2026, and a new SWRO on Pulau Harapan, built by the regency with The Antheia Project, broke ground on 29 July 2026.

Where it fails. Not for want of plants. In May 2020 the Tidung plant stopped because a process chemical ran out; the SDA agency borrowed stock from Kelapa and said it hoped to restart it (the restart itself is not reported). In July 2025 the Kelapa unit was malfunctioning. In 2016 a nine-plant contract reached 3% before its contractor was blacklisted. And in September 2026 the northern group's wells failed: PHE OSES shipped 13,000 L to 112 families on Pulau Kelapa, about 116 L a family, two days at Sphere's 15 L a person. The northern islands (Kelapa, Kelapa Dua, Harapan, Sebira) are short first in the dry season and cut off first in the monsoon: on 23 January 2026 BMKG forecast 1.25–2.5 m in the northern waters against 0.5–1.25 m in the south, and the Kelapa–Sebira leg stayed closed longest.

* Suggested formulation: check with a health worker before sharing.

Water quality: on Pulau Kelapa, 76% of 240 households say their water tastes brackish, and piped, refill, well, and rain water all failed the total-coliform standard (Azni et al., JKLI 2025). Wells on Pramuka were reported saline across the island by 2012, and in 2021 the regency built 266 recharge wells after blaming over-pumping for brackish wells.

WHAT THIS PLACE SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort before and through the peak. Backing lenses are shown per action. Convergence was checked for shared bias. Each action carries the RPB's columns: owner, place, target, date, and source of funds.

1. By 30 November, put a weather-gated referral rule in writing: when BMKG Maritim forecasts waves of 1.25 m or more, move known-risk patients from the outer islands before the sea closes, with a pre-agreed KSOP clearance for ambulance boats

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY · FRONTLINE | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
Sudin Kesehatan Kepulauan Seribu, with RSUD Kepulauan Seribu, PK3D Dinkes DKI, and the two puskesmas; KSOP Muara Angke and Dishub consulted	**Rule and patient list by 30 Nov 2026; drill in December; in force to March 2027**	**Staff time and existing boats; not costed**	**A dengue, obstetric, or dialysis patient stranded on an outer island when the boats stop**

RSUD Kepulauan Seribu on Pulau Pramuka is the only referral hospital, with a 24-hour emergency department; anything it cannot handle goes to the mainland (a Sebira patient was referred to RSUD Koja). Five ambulance boats were on standby in June 2026: two of PK3D's (at Tidung and Kelapa), two of the RSUD's (at Pramuka), and one of Puskesmas Seribu Utara's (at Kelapa). No published rule says what they do when KSOP stops sailing. The rule is simple: when BMKG Maritim forecasts 1.25 m or more for Perairan Kepulauan Seribu (Bagian Utara or Selatan), the puskesmas moves its list of known-risk patients (late pregnancy, dialysis, dengue with warning signs) to Pramuka or the mainland in the last safe window, and KSOP has agreed in advance how a medical run is cleared. In January 2026 the boats stopped from 12 to 14 January and again on 28 January at exactly that wave band. Kemenkes reports that Indonesia's dengue deaths in 2025 (0.42% of cases) came mainly from patients reaching care late.

* Suggested formulation: check with a health worker before sharing.

Corrections: the ambulance boats are described as 24/7, but their wave limits are not published; the RSUD's class and the completion of its ICU renovation (78% in February 2026) are unverified. In December 2021 the coordinating ministry said that Puskesmas Tidung had only two inpatient rooms for about 5,000 people and that "many patients ended up not being saved" on the long transfer; no number was given. The patient list stays with the puskesmas, not the RT. Province rule: action JK-06 (the clinical alert).

2. From 15 November, PAM Jaya publishes a weekly list of which desalination plant is running on which island, and by 30 November puts spare membranes, pump parts, process chemicals, and generator fuel on each island

PARTIALLY CONFIRMED | EMERGENCY · SKEPTIC · ECONOMIST | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
PAM Jaya, with Sudin SDA Kepulauan Seribu and the regency	First list 15 Nov 2026; spares on every island by 30 Nov; weekly to March 2027	PAM Jaya's own operating budget; not costed (no spare-part price is published)	A plant failing in January and staying down until the boats sail again

Every documented failure here is a consumable, a part, or a repair that did not arrive: the chemical at Tidung in 2020, the unit at Kelapa in 2025. In the monsoon a missing part cannot come by boat, so it has to be on the island before December. A weekly public list, plant by plant, also tells the regency where to send boat water before wells fail, which nobody can see today. In August 2026 the regency, PAM Jaya, and Sudin SDA said they were speeding up maintenance and component replacement; this action dates and publishes it.

* Suggested formulation: check with a health worker before sharing.

Corrections: "fully shifted to PAM Jaya" is attributed to the Sudin SDA head, Mustajab, in Metro TV's report (an earlier draft called it the reporter's own summary). The plant capacities are from 2021, and their current status is unpublished; whether the Kelapa unit was repaired after July 2025 is unknown. Plants: SWRO-KELAPA, SWRO-PANGGANG, SWRO-TIDUNG, SWRO-LANCANG, SWRO-PRAMUKA, SWRO-PAYUNG, SWRO-KELAPADUA, SWRO-UNTUNGJAWA (only Tidung and Panggang are located in the GIS layer).

3. Before 1 December, give a lidded 160 L drum to each of the 112 families on Pulau Kelapa whose wells dried, check them weekly with jumantik, and sign a standby letter with PHE OSES for boat water to the northern islands

PARTIALLY CONFIRMED | EMERGENCY · PHILANTHROPIST · PUBLIC HEALTH | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
Pemkab Kepulauan Seribu (the Bupati's office), with PHE OSES, Kelurahan Pulau Kelapa and the PKK, and jumantik cadres	**Drums and letter by 30 Nov 2026; weekly checks to March 2027**	**Rp22.2 m for 112 drums (F01); the letter costs nothing**	**Households running dry when boats stop, and open drums breeding dengue in January**

The one water delivery on record this season went to these 112 families, from PHE OSES's Pabelokan island, on 26 September. A lidded drum at about Rp198,000 (the province's shared unit cost) holds about two days of water for a family of five at 15 L a person, so it bridges the gap between a delivery and a plant restart, and with its lid on and a weekly scrub it is not a breeding site when the rains come (province action JK-02). A standby letter turns one act of goodwill into an arrangement the regency can call on for Kelapa, Harapan, Kelapa Dua, and Sebira when wells fail or a plant stops.

* Suggested formulation: check with a health worker before sharing.

Corrections: the 112 are families (KK) on Pulau Kelapa only; Pulau Harapan was not on that list, and no count of households without a piped connection exists for any island, so the extension to the other northern islands waits for a kelurahan count (ceiling only). PHE OSES has no recorded co-funding with a donor, so the matching grant is untested. Prices are undated retail listings: release money against a supplier quote.

4. By 1 December, both puskesmas and the pustu triage dengue warning signs and hold rapid tests and IV fluids, so a patient is stabilised before the crossing; after a site check, each gets two closed 500 L tanks of water

PARTIALLY CONFIRMED | PUBLIC HEALTH · PHILANTHROPIST | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
Sudin Kesehatan Kepulauan Seribu; Puskesmas Kepulauan Seribu Utara and Selatan; pustu Untung Jawa, Lancang, Pari, Panggang, and Harapan	1 Dec 2026	Tests and fluids in existing stock lines; Rp20.4 m for 14 tanks (F02), after a site check	A dengue patient deteriorating on an island with no hospital

Selatan's dengue incidence in 2025, 118.54 per 100,000, is higher than four of North Jakarta's six kecamatan. On an island, the place where a warning sign is first seen is the place where the patient must be kept stable, sometimes for a day, until a boat can go. Use the Kemenkes clinical guidelines (PNPK): recognise the warning signs, check haematocrit, give fluids, and refer under action 1. A clinic without water cannot do this, so each facility gets two closed 500 L tanks filled before the monsoon, if a site check shows it has no reserve.*

* Suggested formulation: check with a health worker before sharing.

Corrections: what each pustu can actually do (staff, beds, laboratory) is unpublished, and Sebira and Kelapa Dua have no pustu on the Dinkes list. No dengue outbreak or death on a single island was found for 2023–2026. A gift of tanks to a DKI office is a hibah to the Pemprov (agreement and asset record), so F02 goes through the regency. Province rule: action JK-06.

5. By 30 November, stock the island food outlets with two weeks of rice and cooking oil, three weeks earlier than last year, and list fisher households for JK-09 before the lean season

PARTIALLY CONFIRMED | FOOD SUPPLY · FRONTLINE · EMERGENCY | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
Sudin KPKP Kepulauan Seribu, with Pasar Jaya (JakGrosir Pulau Tidung) and Dishub for the boats	**Stock by 30 Nov 2026; fisher list to the province before 1 Dec**	**Existing province and BUMD lines; not costed**	**Shortages and price spikes during sailing bans, in the months fishers cannot fish**

The province sent 15 t of food to the islands on 11 December 2025 (3 t rice, 3 t sugar, 7 t cooking oil, 1 t frozen chicken, and 1 t of biscuits and syrup, on three boats from Marina Ancol), in the same week in which the boats had stopped for three days the year before (10–12 December 2024). This season, send it by 30 November. Pasar Jaya's JakGrosir on Pulau Tidung Kecil, open since September 2019, sells more than half its goods to KJP subsidy holders. Fishers stay in port from about December to mid-February, which is when the province's rice aid ends and when a rice price forecast to rise through March 2027 bites; fishers are not yet one of the groups eligible for subsidised food, so this action does not register them: the regency lists fisher households for the province, which is asked to make them eligible (action JK-09).

* Suggested formulation: check with a health worker before sharing.

Corrections: LPG was dropped from the "same as last year" framing; the 2025 shipment carried none, and no island LPG shortage was found. No fixed food outlet other than JakGrosir Tidung was confirmed (food bazaars ran on Tidung, Pramuka, Kelapa, Kelapa Dua, and Harapan in March 2025). No dated food shortage during a sailing ban was found, and the island price gap is unpublished. BPBD pre-stocked water and biscuits in all six kelurahan on 12 February 2026, a precedent for the timing.

6. From November, send the December–January rob tide dates to the RT heads on Panggang, Harapan, and Untung Jawa, and keep clinic medicines above floor level there

CONFIRMED | EMERGENCY · HISTORIAN | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
BPBD DKI, through the kelurahan and RT; the puskesmas and pustu	30 Nov 2026	Staff time	Damage to homes, medicines, and the cold chain at king tides

Rob reached 40 cm in six RT of Pulau Panggang at the new moon on 16 December 2024, when BPBD recorded flooding in five of the six kelurahan, and 10–15 cm on Untung Jawa and about 10 cm on Harapan on 4 December 2025 (10 RT in all). BPBD also recorded inundation in all six kelurahan on 3–6 June 2020. None of these killed anyone. This is a cheap action for a damaging, not deadly, hazard, which is why it ranks sixth.

7. Coral heat stress: stop anchoring on snorkel and dive sites now, and survey bleaching if NOAA Coral Reef Watch reaches Warning

PARTIALLY CONFIRMED | FRONTLINE · FOOD SUPPLY | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
Balai Taman Nasional Kepulauan Seribu, with Sudin KPKP, tour operators, and fisher groups	Anchoring ban announced by 30 Nov 2026; survey on Warning	Low; not costed	Loss of fish habitat and dive income after the season

* Suggested formulation: check with a health worker before sharing.

After the 2023–24 El Niño, an IPB team found about half the corals bleached across a 7 ha survey, worst at Gosong Pramuka, and anchor damage at snorkel sites made it worse. NOAA's nearest 5 km station, "Northern Java", has been at Bleaching Watch since 10 October 2026 and reached Alert Level 1 in 2016, 2023, and 2024. This protects incomes, not lives this season, which is why it ranks last.

Corrections: the station is centred about 300 km east of Seribu and its coverage of the islands is unconfirmed; the IPB survey dates are not in the source. Status on 14 October was Watch, not Warning.

WHAT THE PROVINCE DOES FOR KEPULAUAN SERIBU

WRITTEN ONCE

These items are province-wide, so they are written once in the province file (`jakarta-province-*`, code JK) and not priced here.

- **JK-01, the decree.** BMKG's Awas trigger has fired for Kepulauan Seribu, but DKI has declared no status. A Keputusan Gubernur declaring siaga darurat kekeringan lets BTT be shifted (by pergeseran) into water lines and opens a request for BNPB Dana Siap Pakai (Perban BNPB 4/2020); only tanggap darurat opens BTT's direct release.
- **JK-02**, "tutup wadah air, kuras dan sikat seminggu sekali" on every notice and delivery. **JK-05, trigger cash:** Rp300,000 once per household per declared status, at most twice a season, through Bank DKI, only after a written BPKD and Inspektorat opinion. Seribu promises no cash of its own.
- **JK-06**, the dengue and leptospirosis clinical alert; **JK-09**, SPHP rice outlets and fishers added to subsidised food; **JK-10 and JK-11**, asks to central government for January–March rice aid and Dana Siap Pakai.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Pemkab (Bupati)	Sudin Kesehatan KS	RSUD KS	PK3D Dinkes	Puskesmas & pustu	KSOP Muara Angke	Dishub DKI	PAM Jaya	Sudin SDA KS	PHE OSES	Kelurahan, RT & PKK	Jumantik	Sudin KPKP KS	Pasar Jaya	BPBD DKI	Balai TN
1. Weather-gated referral	I	A	R	R	R	C	C									
2. Plant status list and spares	C							A	R		I					
3. Kelapa drums, PHE OSES letter	A	C							C	R	R	R				
4. Dengue triage, clinic water		A	C	I	R											
5. Food pre-stock, fishers	I						R				C		A	R		
6. Rob tide dates					R				C		R				A	
7. Reef anchoring ban											C		R			A

What the chart shows: the health office carries the two highest-ranked actions, and one meeting between Sudin Kesehatan, the RSUD, PK3D, and KSOP settles action 1. PAM Jaya alone answers for whether the plants run. Only action 3 needs outside money, and only action 4 needs a gift in kind.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

Rp22.2 m (~US$1,236): drums for the 112 Kelapa families

TARGET

Rp42.6 m (~US$2,374)

CEILING

Rp119.4 m (~US$6,655)

LARGEST UNCERTAINTY

How many island households have no piped connection; no count is published

Floor = 112 × Rp198,000 = Rp22.176 m. Target = Rp22.176 m + Rp20.44 m (14 tanks at 7 health facilities) = Rp42.616 m. Ceiling = 500 families × Rp198,000 = Rp99 m (an assumption, pending a kelurahan count) + Rp20.44 m = Rp119.44 m. Rate: JISDOR Rp17,948 per US$ (15 October 2026). **All amounts are readiness money; none is committed.** The target fell 94% in verification, from the panel's Rp656.5 m: a 1,000-household drum target with no denominator, Rp325.6 m of community larvicide rounds sized on a household count derived by division (and in tension with the province rule on temephos, JK-X02), and Rp100.8 m of fishers' BPJS premiums (income protection, not lives; the rate's coverage is disputed) were cut. Not counted: the province trigger-cash line JK-F01, and water from PAM Jaya at its subsidised tariff. **Public**

* Suggested formulation: check with a health worker before sharing.

money: the 2027 APBD ceilings were fixed when the KUA-PPAS was signed on 13 August 2026, and the RAPBD 2027 (Rp82.89 tn) was tabled on 13 October, with joint approval due about 30 November. None of these actions is a new APBD 2027 line. The regency has no budget of its own; public money can come only by reallocation within existing dinas and suku dinas lines during the deliberations, from the 2026 BTT on a declared status (by pergeseran under siaga darurat), from the APBD-P 2027, or from BNPB Dana Siap Pakai.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
PHE OSES as matching partner, through Kelurahan Pulau Kelapa and the PKK	Lidded 160 L drums for the 112 families whose wells dried (action 3); other northern islands only after a count	Rp22.2 m / 22.2 m / 99 m	Matching grant	112 × Rp198,000 (undated retail); ceiling 500 families, an assumption. PHE OSES is the only body with a recorded water delivery on the islands this season; no co-funding precedent	Partially confirmed
Sudin Kesehatan Kepulauan Seribu, as goods in kind through the regency	Two closed 500 L tanks at each of 2 puskesmas and 5 pustu (action 4), after a site check	Rp20.4 m (target)	Pre-positioned stock	7 × 2 × Rp1.46 m (province unit cost). Existing reserves unpublished	Unverified

The funder's one thing: almost nothing here needs a donor. The plants are PAM Jaya's to run, the boats are the government's, and the referral rule costs a meeting. The money that matters is a few spare membranes and a jerrycan of antiscalant on the right island in November, and that sits in PAM Jaya's own budget. A donor's best use is the one gap no public line covers: lidded storage for the families whose wells have already failed.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives, money, or the window here. Cards 1 to 4 are province-wide (written once as JK-X01 to JK-X04); cards 5 to 8 are Kepulauan Seribu's own, and cards 6 and 8 keep their island text under the province cards JK-X08 (wells) and JK-

* Suggested formulation: check with a health worker before sharing.

X09 (relocation).

Do not stand the response down when the rains return

CONTRAINDICATED

The rains bring the west monsoon, and the monsoon closes the sea. Water stored for the drought becomes breeding stock within one to two weeks, and DKI's dengue peaked in mid-March 2004 and in April 2024.

Instead: keep the referral rule, plant stocks, and weekly checks running to March 2027; stand down on falling dengue cases and calm seas, not on rainfall.

Do not make fogging the visible dengue response, and do not bulk-buy temephos without a bioassay

CONTRAINDICATED

Jakarta *Aedes aegypti* resist permethrin, the fogging class (Hamid et al., PLoS ONE 2017). In Camino Verde, households using temephos had higher odds of child infection (aOR 1.44). No Seribu bioassay exists.

Instead: fog only on a case investigation; put the effort into lids and weekly scrubbing (action 3); choose any larvicide after a WHO bioassay.

Do not tell households to "store water" without "cover it, drain and scrub it weekly"

CONTRAINDICATED

Aedes eggs stick to container walls and survive draining. On islands where every house keeps drums, rain tanks, and toren, storage advice alone multiplies breeding sites.

* Suggested formulation: check with a health worker before sharing.

Instead: every delivery and notice carries the whole line (province action JK-02).

Do not present boat-delivered, tanker, well, or refill-depot water as drinking water*

CONTRAINDICATED

On Pulau Kelapa, piped, refill, well, and rain water all failed the total-coliform standard, and well and rain water failed for *E. coli* (Azni et al., JKLI 2025). Water shipped in bulk, by boat or tanker, sits in open containers for days.

Instead: label it "air bersih, bukan air minum: rebus dulu"; drink boiled or gallon water.

Do not buy new desalination units, or bottled water as a standing supply, to cover a plant that is down

CONTRAINDICATED

Seribu's plants have failed for want of a chemical (Tidung, 2020) or a repair (Kelapa, 2025), not for want of plants, and a 2016 build contract stalled at 3%. A new unit cannot be built before the monsoon; bottled water bought in place of a repair takes the pressure off fixing it.

Instead: spares and chemicals on each island and a weekly status list (action 2); bulk water by boat is a bridge while a plant restarts, not a substitute.

Do not drill or deepen wells, or add pumps, as the drought response on the islands

CONTRAINDICATED

A small island's fresh water floats on seawater. In 2021 the regency blamed over-pumping for brackish and salty wells and built 266 recharge wells; on Pramuka, wells were reported saline across the island by 2012 (a field observation, not a measurement).

* Suggested formulation: check with a health worker before sharing.

Instead: keep wells for washing; route drinking water through the plants; ask Sudin SDA to measure well salinity before and after the season.*

Do not sail aid or overload a boat when KSOP has withheld sailing permits

CONTRAINDICATED

The Zahro Express burned 15 minutes out of Muara Angke on 1 January 2017 and at least 23 people died; its permit listed 100 passengers and 247 were aboard (Tirto). KSOP withheld permits on 14 January 2026 and the fast boats stopped again on 28 January in 1.25–2.5 m waves.

Instead: pre-stock water, medicine, and food by 30 November (actions 2 to 5), and move patients early under a cleared medical run (action 1).

Do not use disaster risk, seawalls, or setback lines to justify relocating residents on Pulau Pari

CONTRAINDICATED

A company claims about 90% of Pulau Pari under 62 freehold and 14 building-use certificates whose issuance the Ombudsman found maladministered in 2018; three residents were arrested in 2017 and acquitted on appeal in 2018, and BPN has still made no ownership decision (CNA, 24 August 2026). In January 2025 the fisheries ministry sealed unpermitted dredging and fill by another company nearby. Risk language here will be read as taking a side.

Instead: protect residents where they live, work through the RT and the residents' forum alongside the kelurahan, and keep every action off disputed land.

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1 Jan 2017	Zahro Express fire off Muara Angke, on the Tidung route; 247 aboard against a permit for 100 (neutral)	At least 23 dead (BNPB; later counts 24)	Harbourmaster dismissed
May 2020	Tidung SWRO stops: process chemical runs out (neutral)	No deaths reported	Chemical borrowed from Kelapa; restart not reported
3–6 Jun 2020	Inundation of 5–50 cm in all six kelurahan (BPBD record)	No deaths	Nothing found
2023–24	Coral bleaching after the El Niño, about half of a 7 ha survey (IPB), reported May 2024 (neutral by then)	Habitat and income loss	Nothing found
10–16 Dec 2024	Boats suspended three days on BMKG advice; rob up to 40 cm on Panggang, flooding in five kelurahan (neutral)	No deaths	Seawalls on Lancang, Tidung, Panggang completed
Jul 2025	Kelapa seawater unit malfunctioning; bathing water short	No deaths reported	Repair ordered; completion unreported
Dec 2025	Rob on Untung Jawa and Harapan (10 RT); 15 t of food shipped on 11 Dec	No deaths	Nothing specific
Jan 2026	Boats stopped 12–14 Jan and again 28 Jan, waves 1.25–2.5 m, no KSOP permits (neutral)	No deaths reported	BPBD pre-stocked water and biscuits in six kelurahan (12 Feb)
Sep 2026	El Niño dry season: Awas drought warning; Kelapa's wells dry or brackish for about six weeks	No deaths reported	PHE OSES shipped 13,000 L to 112 families; no government delivery found

The recurrence nobody has noticed: the islands' water and the islands' boats fail in the same place, in the same order. The northern group is short first in the dry season and cut off first in the monsoon, and every documented plant failure was a small thing that could not reach the island in time. No death from water shortage or sea flooding was found; the deaths on record are at sea. No record was found of water crises or

* Suggested formulation: check with a health worker before sharing.

deliveries in the 1997, 2015, or 2023 El Niño dry seasons. ENSO phases follow NOAA's ONI (ERSSTv6, five-season rule). City-wide events (1996, 2002, and 2007 floods; 1997–98 rice; dengue in 1998, 2004, and 2024) are in the province file.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Fund community larvicide rounds in both kecamatan** (Philanthropist, Rp325.6 m) versus **do not bulk-buy temephos without a bioassay** (Skeptic; province card JK-X02). Lids and weekly scrubbing were kept; a WHO bioassay on island larvae settles the rest.
- **Insure fishers through BPJS Ketenagakerjaan before the monsoon** (Economist, after Kepri's Rp7.3 bn for 36,500 fishers) versus **it protects income, not lives, this season** (the funding verifier). Left out of the envelope; a policy for the regency's next budget.
- **Seawalls protect low islands** (the regency built 340–510 m on Lancang, Tidung, and Panggang in December 2024) versus **they move abrasion along the shore and, on Pari, read as taking a side in a land dispute** (Skeptic). No effects study was found.

WHAT NOBODY COVERED — BLIND SPOT

Which plant is running today, and who has no pipe. No source after 2021 says which of the islands' desalination plants work, at what output, and no source counts the households that rely on wells, rain tanks, or bought water. Both numbers sit inside PAM Jaya and the kelurahan offices. Until they are published, this briefing cannot say whether Kelapa's September shortage was a plant failure, a capacity gap, or wells alone, and the funding ceiling stays an assumption. The Pemkab Kepulauan Seribu letter in the set's data requests asks for both.

HOW TO READ THIS — METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale. Here the only row is the whole province (Jakarta Raya, 10.7 million people), so Seribu's figure is a population share, and at under one death it says only that heat is not this place's hazard. The El Niño projection covers *heat only*; water, the sea, and dengue are carried here by the historical record rather than by a published number.

* Suggested formulation: check with a health worker before sharing.

- **Scope.** The location was taken as given: Kabupaten Administrasi Kepulauan Seribu, DKI Jakarta (Kemendagri 31.01; 2 kecamatan, 6 kelurahan; 29,355 people, BPS 2026 projection). Codes come from the GIS `kelurahan` layer's Kemendagri field (BPS numbers the two kecamatan the other way round). Climate Impact Lab impact region: GADM IDN.7_1 "Jakarta Raya", the whole province. Its row (about 260 excess heat deaths, September 2026 – February 2027; 147 in December–February) was scaled by population share (0.28%) to about 0.7, our arithmetic. Kelurahan is the finest unit named; no RW is ranked; dengue is shown no finer than kecamatan.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts applied to temperature–mortality relationships, against a 1996–2025 baseline. Country and province figures cover September 2026 – February 2027. Heat only.
- **Expert panel.** The eight lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) were run as five research agents, pairing emergency with historian, frontline with food, and economist with philanthropist, each to a fixed brief carrying the set's verified facts (PLAN.md log). The province items were not re-researched.
- **Local evidence.** From the project GIS base (`projects/elnino-jakarta/gis/jakarta_base.gpkg`), built from Satu Data Jakarta: BPBD flood events by kelurahan (2013–2024), Dinkes dengue by kecamatan (2024 suspected cases, 2025 incidence), puskesmas and pustu locations, and the two located desalination plants.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias: three lenses converged on "stock the islands before the boats stop", which rests on local records (the January 2026 suspensions, the September Kelapa delivery) rather than on shared literature. Contradictions were kept. Actions were ranked by expected lives saved per unit of effort, each was given a payer or marked as staff time, and the placements were reconciled against the ranked list.
- **Verification.** Three verifier agents checked all 15 recommendations, the funding envelope, and the record: water and boats; health, food, and reefs; and funding and the Do not cards (with the financial checks). Their corrections are shown under each action.
- **Who does what.** The RACI chart is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not searchable this run (it renders no records until a country filter is applied in a browser); unsearched, not "none exists".

* Suggested formulation: check with a health worker before sharing.

- **Health wording.** Passages where a wording error could hurt someone (dengue danger signs and when to seek care, and drinking-water safety) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented. One exchange rate for the set: JISDOR Rp17,948 per US$ (Bank Indonesia, 15 October 2026). Every changing reading carries its date. Indonesian text for every action, funding line, and event is in `kepulauan-seribu-id.json`, not yet reviewed by a native speaker.
- **Revised 2026-10-15 after the set audit** (fixes applied: ENSO labels for January 2017, May 2024, and December 2024 set to neutral under ONI v6; fishers listed for JK-09, not registered, for subsidised food; Puskesmas Tidung's two inpatient rooms, not beds; the Tidung 2020 restart marked unreported; "fully shifted to PAM Jaya" attributed to Mustajab; the December 2025 shipment itemised to 15 t; the unlinked Republika dengue count dropped; BMKG's *Awas* lists for September and October cited as the set records them; the coral trigger caveat restored in Indonesian; do-not cards 6 and 8 referenced to JK-X08 and JK-X09, card 4 widened to well and refill-depot water; UK spelling; the Indonesian premise added to the language pack). The envelope is unchanged.
- **Limits.** The panel and verifiers are AI research agents working from web sources; no one on the islands was interviewed. Several RRI and Koran Jakarta pages returned 403 and were read only as search snippets (the August 2026 maintenance report, the RSUD renovation, the PAM Jaya tariff). Run-specific gaps: plant status after 2021; households without a piped connection; pustu capability; ambulance-boat wave limits; island food and LPG prices; the coverage of NOAA's Northern Java station; water history in earlier El Niño years; whether any DKI status has been declared since September.

References · Water: supply and need, Oct 2025, SDA budget and PAM Jaya handover (Mustajab), Nov 2025, 2026 BWRO plans, Pergub 62/2019, plant capacities 2021, tariff 2021, Tidung 2020, Kelapa 2025, 2016 contract, Harapan SWRO, maintenance, Aug 2026 (snippet), PHE OSES delivery, Kelapa water quality, recharge wells, Pramuka 2012, Sphere (confirmed, corrected) · Sea and boats: Jan 2026 stop, BMKG forecast, 28 Jan 2026, Dec 2024, BMKG Maritim, Zahro Express, Zahro Express permit, rob Dec 2024, rob Dec 2025, seawalls 2024, Satu Data Jakarta (BPBD, Dinkes tables) (confirmed, corrected) · Health: ambulance boats, RSUD, RSUD renovation, Tidung inpatient rooms, 2021, Kemenkes dengue CFR, PNPK dengue, BPS 2024 ages, Riskesdas 2018, Camino Verde, Hamid 2017 (confirmed, corrected) · Food and reefs: 15 t, Dec 2025, JakGrosir Tidung, JakGrosir sales, BPBD pre-stock 2026, NOAA CRW Northern Java, bleaching 2024 (confirmed, corrected) · Pari and money: Pari land dispute, KKP seal, PHE OSES programmes, Kepri fishers, KUA-PPAS 2027, Perban BNPB 4/2020, drum price, tank price, JISDOR (confirmed, corrected) · Season: BMKG Awas, Sep 2026, BMKG Awas, Oct 2026 (bmkg.go.id), NOAA ONI v6
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

* Suggested formulation: check with a health worker before sharing.

Generated 15 October 2026 for Kabupaten Administrasi Kepulauan Seribu, DKI Jakarta, Indonesia. Indicative, not for land or permit decisions. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

* Suggested formulation: check with a health worker before sharing.